

Here is Claude's plan:

----- IFA-AI — Interactivity,
Icons & Demo Data Polish PR

Context

The IFA frontend (SvelteKit 2 / Svelte 5 / Tailwind, lucide-svelte) looks finished but is largely a static shell. Three concrete problems prompted this work:

1. Broken "Continue" flow. The home "Continue Learning" button (routes/+page.svelte:24 handleContinueLearning) only fires a toast; it never navigates. There is no course-detail/lesson route at all, so no Continue button anywhere can lead to real content. Every "Continue" handler across pages is a console.log no-op.
2. AI-generated-looking, repetitive icons. Sparkles is reused as the section badge in ~8 places (recommendations, activity, continue, research...) and Star labels non-rating things (e.g. "Skills"). The user wants specific icons per section from the already-installed lucide-svelte (e.g. Lightbulb for recommendations/ideas).
3. Dead buttons & disconnected pages. Many buttons have no handler (Header bell/theme/profile, "Enroll Now", peer Share/Invite, topic-remove) and several nav links 404 (/public-courses, /ai-tutor, /recommendations). Data is duplicated and divergent across pages, and two filters are broken so grids render empty.

Outcome: a self-contained frontend-only PR that makes the whole app feel live and connected — a real course→module→lesson→quiz flow driven by a single demo-data store, every button wired, and a coherent per-section icon system — so the product can be demoed convincingly before the real AI pipeline lands.

ScholarXIV branch review (the user's other question)

origin/feature/pr-02-3-real-scholarxiv-integration = the current pr-06 UI plus 9 backend commits (real ScholarXIV HttpClient integration, LearnerProfile/ResearchPackage entities, .NET 10 bump). It changes zero client/ files.

- Integration quality: solid — typed IHttpClientFactory client, graceful degradation, coherent DI.
- Verdict: acceptable to merge with caveats — one migration drops existing Quiz/Module/Lesson completion columns (data loss on migrate), it forces the whole solution/CI onto .NET 10, can break other branches referencing the removed entity fields, and a fresh clone's first /research call 500s unless SCHOLARXIV_API_KEY_IFA is set. Low risk for a throwaway hackathon DB.
- Decision reflected here: this new PR is based on feature/pr-06-hackathon-polish, independent of ScholarXIV (no conflicts either way). Merging ScholarXIV is a separate backend decision.

User decisions (captured)

- Base branch: off feature/pr-06-hackathon-polish.
- Course depth: full lessons + live progress + an interactive demo quiz.
- Wiring scope: everything (course flow and Header extras / orphaned components).

Approach

Introduce a single source of truth for courses and a central icon registry, add the missing course-detail/lesson/quiz route, then point every page and button at these. Reuse the existing string→component icon pattern (LearningOverviewDonut.svelte:5-11) and the already-built-but-orphaned components (NotificationCenter, CommandPalette, LearnerProfileMenu, RecommendationEngine, CourseShareDialog, HeroGoalSuggestions, ActivityStream).

New files

- client/src/lib/data/catalog.ts — the demo data: rich Course[] with modules → lessons (with reader content) → quiz (questions/answers), plus marketplace catalog. Typed via new interfaces.
- client/src/lib/stores/coursesStore.ts — writable enrolled Course[] + actions enrollCourse, toggleBookmark, markLessonComplete, resumeCourse, pauseCourse; derived activeCourses, completedCourses, pausedCourses, bookmarkedCourses, activeCourse, overallProgress. Progress = completed lessons / total, recomputed reactively so "mark complete" updates the donut, my-learning, courses, and home together.
- client/src/lib/stores/uiStore.ts — open state for command palette, notifications, profile menu; theme ('light'|'dark') with localStorage persistence + document.documentElement class toggle.
- client/src/lib/icons.ts — central registry: skillIcons, activityIcons, recommendationIcons, getIcon(name). Consolidates the duplicated maps.
- client/src/routes/courses/[id]/+page.svelte — course detail: module/lesson accordion, mark-complete, progress header, "Take Quiz".
- client/src/lib/components/LessonView.svelte — lesson reader (content + Mark Complete + Next).
- client/src/lib/components/QuizView.svelte — interactive quiz: pick answers, submit, score, per-question explanation, updates progress.
- client/src/routes/recommendations/+page.svelte — wraps orphaned RecommendationEngine.
- client/src/routes/ai-tutor/+page.svelte — full-page tutor (reuses AiTutorDock internals / chat UI).

Extend types — client/src/lib/types/index.ts

Add Lesson, Quiz, QuizQuestion, Module, Course (status 'inProgress' | 'completed' | 'paused' | 'notStarted'). Type the previously-any page arrays against these.

Fix the Continue flow & broken filters

- routes/+page.svelte:24 → goto('/courses/' + \$activeCourse.id); wire handleSelectRecommendation to goto(rec.actionUrl); make Enroll add to store.
- ContinueLearningCard.svelte, CourseLibrary.svelte, my-learning, my-skills Continue/Resume/Review buttons → goto('/courses/[id]') (or lesson).
- Bug: routes/courses/+page.svelte:70 'in-progress' → 'inProgress' (grid currently always empty while header counts 3).
- Bug: routes/marketplace/+page.svelte:126 category compare fails for multi-word// categories — normalize both sides (strip non-alphanumerics, lowercase); the "Backend" chip has no matching course (add one or drop the chip). Also swap dead via.placeholder.com thumbnails for reliable ones.
- Read course data on courses, my-learning from coursesStore instead of local arrays; make bookmark toggles reactive (current courses/+page.svelte:91 mutation doesn't reassign).

Wire dead buttons & connect pages

- Header.svelte — bell → NotificationCenter; search + global ⌘K → CommandPalette; profile block → LearnerProfileMenu; Sun → theme toggle (via uiStore). Replace console.log in CommandPalette:31 / NotificationCenter:69 / LearnerProfileMenu:31 with real goto.

- Sidebar.svelte — /public-courses → /marketplace; /ai-tutor → new page; brand icon (Sparkles→GraduationCap).
- Share buttons (ContinueLearningCard, ActiveCourseCard) → open CourseShareDialog instead of toast.
- PublicCoursesGrid.svelte:109 Enroll, marketplace enroll/bookmark, PeerComparisonView Share/Invite, ResourcePreferences add/remove topic, my-skills/progress/research console.log handlers → real actions (enroll→store+navigate, others→toast/state).
- Surface orphans in real homes: HeroGoalSuggestions under the hero; ActivityStream on /progress; RecommendationEngine on /recommendations.
- Surface orphans in real homes: HeroGoalSuggestions under the hero; ActivityStream on /progress; RecommendationEngine on /recommendations.

Icons — specific per section (from lucide-svelte)

Route section-badge icons through intent, not decoration:

- Recommendations / ideas → Lightbulb (RecommendationsList:13, RecommendationEngine:47).
- Recent activity → History/Activity; Continue Learning → PlayCircle; Research → FlaskConical; Goals → Target; brand → GraduationCap.
- Non-rating Stars: LearnerProfileMenu:19 (Skills → Target), ActivityStream:32 (achievement → Award). Keep genuine rating stars (PublicCourseCard:47, PublicCoursesGrid:96) and AI-orb Sparkles (AiTutorDock, HeroSection).
- Per-item recommendation icons via recommendationIcons (review→RefreshCw, continue→PlayCircle, practice→Target). Remove unused Star imports (marketplace:2, PeerProgressComparison:2).

Theme toggle (largest / lowest-priority item)

Make light/dark real: convert tailwind.config.js ifa colors to rgb(var(--ifa-*) / <alpha-value>) (preserves the many /10–/30 opacity utilities), define light :root + .dark channel triples in app.css, drive via uiStore.theme from both Header Sun button and Settings ThemePreferences. Implemented last so it can't block the core flow; can be de-scoped to "toggle + persist only" if dark visuals need more time.

Critical files to modify (representative)

- Stores/data/types: lib/stores/dashboardStore.ts, new coursesStore.ts/uiStore.ts/data/catalog.ts/icons.ts, lib/types/index.ts
- Routes: routes/+page.svelte, routes/courses/+page.svelte, new routes/courses/[id]/+page.svelte, routes/my-learning/+page.svelte, routes/marketplace/+page.svelte, routes/my-skills/+page.svelte, routes/progress/+page.svelte, routes/research/+page.svelte, new routes/recommendations/+page.svelte, new routes/ai-tutor/+page.svelte
- Components: Header.svelte, Sidebar.svelte, ContinueLearningCard.svelte, CourseLibrary.svelte, RecommendationsList.svelte, PublicCoursesGrid.svelte, PeerComparisonView.svelte, ResourcePreferences.svelte, RecentActivityFeed.svelte, ResearchSummary.svelte, and the wired orphans (NotificationCenter, CommandPalette, LearnerProfileMenu, RecommendationEngine, CourseShareDialog, HeroGoalSuggestions, ActivityStream), plus new LessonView.svelte/QuizView.svelte
- Theme: tailwind.config.js, app.css

Verification

1. cd client && npm run check (svelte-check: types/props/unused imports clean).
2. npm run dev and walk the demo:
 - Home Continue Learning → lands on /courses/csharp-backend; open a lesson, Mark Complete → progress %, donut, My Learning, and Courses all update together.

- Take a module quiz → answer, submit, see score + explanations; progress reflects it.
 - Courses: "In Progress" filter now shows courses; bookmark toggles live; Continue navigates.
 - Marketplace: category chips (Data Science, AI/ML, DevOps) filter correctly; Enroll adds the course and routes into it; thumbnails load.
 - Header: bell opens notifications; #K / search opens the command palette and navigates; profile menu links work; Sun toggles light/dark and persists on reload.
 - Sidebar: every item resolves (no 404); "Public Courses" → marketplace; AI Tutor + Recommendations pages render.
 - No console.log-only buttons remain in live pages; recommendation/idea sections show Lightbulb, sections show intent-specific icons, rating stars unchanged.
3. npm run build succeeds.
 4. Commit on a new branch off feature/pr-06-hackathon-polish and open the PR.